

$$1 - \operatorname{sech}^2 t = \tanh^2 t. \quad [3]$$

[illegible]
$$x = \frac{1}{2} \tanh^2 t + \ln \operatorname{sech} t, \quad y = 1 + \tanh^4 t, \quad \text{for } t > 0.$$

(b) Show that $\frac{dy}{dx} = -4\operatorname{sech}^2 t$. [5]

[illegible]

- (c) Find the coordinates of the point on C with $\frac{d^2y}{dx^2} = -\frac{9}{2}$, giving your answer in the form $(a + \ln b, c)$ where a, b and c are rational numbers. [6]

[illegible]